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Article of footwear including a sole structure

Abstract

An article of footwear including an upper and a sole structure. The sole structure includes a midsole, wherein the midsole includes an enclosed cavity, a thermoformed cushioning element disposed within the enclosed cavity, wherein the cushioning element includes an upper barrier film and a bottom barrier film enclosing an internal volume, and an outsole layer, wherein the outsole layer forms a ground-engaging surface of the article of footwear.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application claims priority to U.S. Provisional Application No. 63/322,839, filed Mar. 23, 2022, the entirety of which is incorporated herein by reference.

FIELD

(1) The present disclosure relates generally to sole structures for articles of footwear and more particularly to sole structures incorporating a fluid-filled bladder.

BACKGROUND

(2) This section provides background information related to the present disclosure which is not necessarily prior art.

(3) Articles of footwear conventionally include an upper and a sole structure. The upper may be formed from any suitable material(s) to receive, secure, and support a foot on the sole structure. The upper may cooperate with laces, straps, or other fasteners to adjust the fit of the upper around the foot. A bottom portion of the upper, proximate to a bottom surface of the foot, attaches to the sole structure.

(4) Sole structures generally include a layered arrangement extending between a ground surface and the upper. One layer of the sole structure includes an outsole that provides abrasion-resistance and traction with the ground surface. The outsole may be formed from rubber or other materials that impart durability and wear-resistance, as well as enhance traction with the ground surface. Another layer of the sole structure includes a midsole disposed between the outsole and the upper. The midsole provides cushioning for the foot and may be partially formed from a polymer foam material that compresses resiliently under an applied load to cushion the foot by attenuating ground-reaction forces. The midsole may additionally or alternatively incorporate a fluid-filled bladder to increase the durability of the sole structure, as well as to provide cushioning to the foot by compressing resiliently under an applied load to attenuate ground-reaction forces. Sole structures may also include a comfort-enhancing insole or a sockliner located within a void proximate to the bottom portion of the upper and a strobel attached to the upper and disposed between the midsole and the insole or sockliner.

(5) Midsoles employing fluid-filled bladders typically include a bladder formed from two barrier layers of polymer material that are sealed or bonded together. The fluid-filled bladders are pressurized with a fluid such as air, and may incorporate tensile members within the bladder to retain the shape of the bladder when compressed resiliently under applied loads, such as during athletic movements. Generally, bladders are designed with an emphasis on balancing support for the foot and cushioning characteristics that relate to responsiveness as the bladder resiliently compresses under an applied load.

Description

DRAWINGS

(1) The drawings described herein are for illustrative purposes only of selected configurations and are not intended to limit the scope of the present disclosure.

(2) FIG. 1 is a perspective view of an article of footwear;

(3) FIG. 2 is a perspective view of a sole structure of the article of footwear of FIG. 1;

(4) FIG. 3 is an exploded perspective view of the sole structure of the article of footwear of FIG. 1;

(5) FIG. 4A is a perspective view of a cushioning element of the article of footwear of FIG. 1;

(6) FIG. 4B is a rear view of the cushioning element of the article of footwear of FIG. 1;

(7) FIG. 4C is a cross-sectional view of the cushioning element of the article of footwear of FIG. 1;

(8) FIG. 5A is a partial view of the sole structure of FIG. 1;

(9) FIG. 5B is a zoomed in partial view of the sole structure of FIG. 1; and

(10) FIG. 6 is a bottom view of a ground-engaging surface of the article of footwear of FIG. 1.

(11) Corresponding reference numerals indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

(12) Example configurations will now be described more fully with reference to the accompanying drawings. Example configurations are provided so that this disclosure will be thorough, and will fully convey the scope of the disclosure to those of ordinary skill in the art. Specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of configurations of the present disclosure. It will be apparent to those of ordinary skill in the art that specific details need not be employed, that example configurations may be embodied in many different forms, and that the specific details and the example configurations should not be construed to limit the scope of the disclosure.

(13) The terminology used herein is for the purpose of describing particular exemplary configurations only and is not intended to be limiting. As used herein, the singular articles “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. Additional or alternative steps may be employed.

(14) When an element or layer is referred to as being “on,” “engaged to,” “connected to,” “attached to,” or “coupled to” another element or layer, it may be directly on, engaged, connected, attached, or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to,” “directly attached to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(15) The terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections. These elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example configurations.

(16) In the discussion that follows, terms “about,” “approximately,” “substantially,” and the like, when used in describing a numerical value, denote a variation of $\pm 10\%$ of that value, unless specified otherwise.

(17) Upper

(18) Referring to FIG. 1, an article of footwear **10** may include an upper **100** and a sole structure **102**. The article of footwear **10** may be divided into one or more regions. The regions may include a forefoot region **12**, a mid-foot region **14**, and a heel region **16**. The forefoot region **12** may be subdivided into a toe portion **12T** corresponding with phalanges, and a ball portion **12B** associated with metatarsal bones of a foot. The mid-foot region **14** may correspond with an arch area of the foot, and the heel region **16** may correspond with rear portions of the foot, including a calcaneus bone. The footwear **10** may further include an anterior end **18** associated with a forward-most point of the forefoot region, and a posterior end **20** associated with a rearward-most point of the heel

region **16**. Referring to FIG. **6**, a longitudinal axis A.sub.10 of the footwear **10** may extend along a length of the footwear **10** from the anterior end **18** to the posterior end **20**, and may generally divide the footwear **10** into a lateral side **22** and a medial side **24**. Accordingly, the lateral side **22** and the medial side **24** respectively correspond with opposite sides of the footwear **10** and extend through the regions **12**, **14**, **16**.

(19) The upper **100** may include interior surfaces that define an interior void configured to, for example, receive and secure a foot for support on sole structure **102**. The upper **100** may be formed from one or more materials that are stitched or adhesively bonded together to form the interior void. Suitable materials of the upper may include, but are not limited to, mesh, textiles, foam, leather, and synthetic leather. The materials may be selected and located to impart properties of durability, air-permeability, wear-resistance, flexibility, and comfort.

(20) In some examples, the upper **100** may include a strobil having a bottom surface opposing the sole structure **102** and an opposing top surface defining a footbed of the interior void. Stitching or adhesives may secure the strobil to the upper **100**. The footbed may be contoured to conform to a profile of the bottom surface (e.g., plantar) of the foot. Optionally, the upper **100** may also incorporate additional layers such as an insole or sockliner that may be disposed upon the strobil and reside within the interior void of the upper **100** to receive a plantar surface of the foot to enhance the comfort of the article of footwear **10**. An ankle opening **104** in the heel region **16** may provide access to the interior void. For example, the ankle opening **104** may receive a foot to secure the foot within the void and facilitate entry and removal of the foot from and to the interior void. In some examples, one or more fasteners **106** may extend along the upper **100** to adjust a fit of the interior void around the foot and to accommodate entry and removal of the foot therefrom. The fasteners **106** may include laces, straps, cords, hook-and-loop, or any other suitable type of fastener. The upper **100** may include a tongue portion that extends between the interior void and the fasteners.

(21) Sole Structure

(22) With reference to FIG. **2**, the sole structure **102** may include a midsole **202** configured to provide cushioning characteristics to the sole structure **102**, a cushioning element **204** configured to provide cushioning characteristics to the sole structure **102**, and an outsole **206** configured to provide a ground-engaging surface of the article of footwear **10**.

(23) Midsole

(24) Referring to FIGS. **2**, **3**, **5A**, and **5B**, the midsole **202** may extend continuously from the anterior end **18** to the posterior end **20**. The midsole **202** may span width of the article of footwear **10** from the lateral side **24** to the medial side **22**. The midsole **202** may include a foam material (e.g. ethylene-vinyl acetate (EVA), polyurethane, or the like.) The midsole **202** may include a top surface **246**, a lateral face **245**, a bottom surface, an interior cavity **224** (shown in FIG. **3**), a heel portion **226**, a forefoot portion **228**, one or more sidewalls **230**, a first foam **223**, and a second foam **225**. Midsole **202** may also include a medial face **247** opposite the lateral face **245**. The medial face **247** may be substantially similar to the lateral face **245**. The first foam **223** may be an upper inner foam of the midsole **202** while the article of footwear **10** is at rest with the outsole **206** facing a ground surface. The second foam **225** may be a lower inner foam of the midsole **202** while the article of footwear **10** is at rest with the outsole **206** facing a ground surface.

(25) Referring to FIGS. **2** and **3**, the top surface **246** may be a substantially smooth surface. Top surface **246** may extend continuously from the heel region **16** to the forefoot region **12**. As top surface **246** extends from the heel region **16** to the forefoot region **12**, top surface **246** may slope downwards in the mid-foot region **14** when viewed from an exterior of the midsole **202**. On medial side **24** and lateral side **22**, one or more sidewalls **230** may extend upward from the top surface **246**.

(26) The lateral face **245** of the midsole **202** may include a window **232**. Window **232** may be defined by an absence of material and/or comprise a material having a transparent region. The

entirety of window **232** may include the material comprising the transparent region. It is contemplated that window **232** may include a material comprising a translucent region. The entirety of window **232** may include the material comprising the translucent region. Window **232** may extend from a portion of the heel region **16** to a portion of the mid-foot region. Window **232** may have a height ranging from about 8 mm-to about 40 mm. Further, window **232** may have a height ranging from about 16 mm to about 20 mm. In an exemplary embodiment, the height of the window **232** is about 19 mm. While window **232** is transparent, window **232** may allow for a person viewing the exterior of the sole structure **102** to see through the material of window **232** to the interior of the sole structure **102**. More particularly, when viewing through window **232**, a person viewing from the exterior will see upper inner foam **223** and the lower inner foam **225**. The viewer will see through the cushioning element **204** and out through the opposing window **232a**. (27) Referring to FIG. 2, with description of the lateral side of the midsole **202**, the one or more sidewalls **230** may extend from the heel region **16** to the forefoot region **12**. The one or more sidewalls **230** may have a first region **232** present in the heel region **16**. The first region **232** may rise towards an apex **234** in the heel region **16**. The one or more sidewalls **230** may extend from the apex **234** towards a second region **236** present in the forefoot region **12**. The one or more sidewalls **230** may slope downwards in the mid-foot region **14** when the one or more sidewalls **230** are extending from the apex **234** toward the second point **236**, when viewed from an exterior of the midsole **202**. The one or more sidewalls **230** may have disposed thereon a plurality of notches **238**. The plurality of notches **238** may be disposed between the apex **234** and the second point **236**. For example, notches **238** may be present on a portion of the one or more sidewalls **230** present in the mid-foot region **14**. It is contemplated that the portion may be within an area of ball portion **12B** disposed within the forefoot region **12**.

(28) The medial side of the midsole **202** may be substantially similar to the lateral side of the midsole **202**. One or more sidewalls **230** may extend from the heel region **16** to the forefoot region **12**. The one or more sidewalls **230** may have a first region present in the heel region **16**. The first region may rise towards an apex in the heel region **16**. The one or more sidewalls **230** may extend from the apex towards a second region present in the forefoot region **12**. The one or more sidewalls **230** may slope downwards in the mid-foot region **14** when the one or more sidewalls **230** are extending from the apex toward the second point, when viewed from an exterior of the midsole **202**. The one or more sidewalls **230** may have disposed thereon a plurality of notches **238**. The plurality of notches **238** may be disposed between the apex and the second point. For example, notches **238** may be present on a portion of the one or more sidewalls **230** present in the mid-foot region **14**. It is contemplated that the portion may be within an area of ball portion **12B** disposed within the forefoot region **12**.

(29) With reference to FIG. 3, the interior cavity **224** may be present within an interior of the midsole **202**. Interior cavity **224** may be defined by an upper inner foam **223** and a lower inner foam **225**. Upper inner foam **223** and lower inner foam **225** may comprise one or more interior portions of midsole **202**. Upper inner foam **223** may comprise an upper interior portion of midsole **202**. Lower inner foam **225** may comprise a lower interior portion of midsole **202**. Interior cavity **224** may further be defined by the windows **232** and **232a**. Interior cavity **224** may be configured to enclose the cushioning element **204**.

(30) The forefoot portion **228** may be disposed in the forefoot region **12**. The anterior most end **229** of the forefoot portion **228** may be curved radially inward towards the center of midsole **202**. Alternatively, it is contemplated that the anterior most end **229** may curve radially outward.

(31) Referring to FIG. 5A, the heel portion **226** may be disposed at a rearmost portion of the midsole **202** corresponding to the heel region **16** and the posterior end **20**. A bottom surface **234** of the heel portion **226** may taper inward from a first point **235** present in the heel region **16** toward a second point **237** present at the posterior end **20**. A top surface **236** of the heel portion **226** may be substantially flat when viewed from above. The top surface **236** may curve radially outward at its

posterior most portion. The heel portion **226** may have disposed thereon one or more slots **288**. The one or more slots **288** may span an entire width of the heel portion **226** and may extend into a portion of the lateral face **245** and the medial face **247**.

(32) Cushioning Element

(33) With reference to FIGS. **4A-4C** and **5B**, the cushioning element **204** of the midsole **202** may include an opposing pair of barrier films **242** and **244**, which can be joined to each other at discrete locations to define a chamber **246**, a web area **248** (divided into web areas **248a**, **248b**, **248c**, **248d**, and **248e**), and a peripheral seam **250**. Cushioning element **204** may be an airbag or a bladder.

(34) Referring to FIG. **4C**, the barrier films **242** and **244** may include a first, upper barrier film **242** and a second, lower barrier film **244**. Alternatively, the chamber **246** can be produced from any suitable combination of one or more barrier films. Web area **248** may be an area where opposing barrier films **242** and **244** are bonded directly to one another without an intervening gap between the films.

(35) As used herein, the term “barrier film” or “barrier membrane” (e.g., barrier films **242** and **244**) encompasses both monolayer and multilayer films. In some embodiments, one or both of barrier films **242** and **244** may each be produced (e.g., thermoformed or blow molded) from a monolayer film (a single layer). In other embodiments, one or both of barrier films **242** and **244** may each be produced (e.g., thermoformed or blow molded) from a multilayer film (multiple sublayers). The multi-layered film may comprise a plurality of layers. The plurality of layers may comprise one or more barrier layers. The one or more barrier layers may comprise a barrier material. The barrier material may comprise or consist essentially of one or more gas barrier compounds. The multi-layered film may comprise at least 5 layers or at least 10 layers. In other embodiments, the multi-layered film may comprise from about 5 layers to about 400 layers. In one aspect of a multi-layered film, the plurality of layers may include a series of alternating layers, in which the alternating layers include two or more barrier layers. Each of the two or more barrier layers may individually comprise a barrier material, the barrier material comprising or consisting essentially of one or more gas barrier compounds. In the series of alternating layers, adjacent layers may be individually formed of materials which differ from each other at least in their chemical compositions based on the individual components present (e.g., the materials of adjacent layers may differ based on whether or not a gas barrier compound is present, or differ based on class or type of gas barrier compound present), the concentration of the individual components present (e.g., the materials of adjacent layers may differ based on the concentration of a specific type of gas barrier compound present), or may differ based on both the components present and their concentrations.

(36) The plurality of layers of the multi-layered film may include first barrier layers comprising a first barrier material and second barrier layers comprising a second barrier material, wherein the first and second barrier materials differ from each other as described above. The first barrier material may be described as comprising a first gas barrier component consisting of all the gas barrier compounds present in the first barrier material, and the second barrier material may be described as comprising a second barrier material component consisting of all the gas barrier compounds present in the second barrier material. In a first example, the first barrier component may consist only of one or more gas barrier polymers, and the second barrier component may consist only of one or more inorganic gas barrier compounds. In a second example, the first barrier component may consist of a first one or more gas barrier polymers, and the second component may consist of a second one or more gas barrier polymers, wherein the first one or more gas barrier polymers differ from the second one or more gas barrier polymers in polymer class, type, or concentration. In a third example, the first barrier component and the second barrier component both may include the same type of gas barrier compound, but the concentration of the gas barrier compound may differ. Optionally the concentrations may differ by at least 5 weight percent based on the weight of the barrier material. In these multi-layered films, the first barrier layers and the second barrier layers may alternate with each other, or may alternate with additional barrier layers

(e.g., third barrier layers comprising a third barrier material, fourth barrier layers comprising a fourth barrier material, etc., wherein each of the first, second, third and fourth, etc., barrier materials differ from each other as described above).

(37) In either aspect, each layer or sublayer can have a film thickness ranging from about 0.2 micrometers to about 1 millimeter. In further embodiments, the film thickness for each layer or sublayer can range from about 0.5 micrometers to about 500 micrometers. In yet further embodiments, the film thickness for each layer or sublayer can range from about 1 micrometer to about 100 micrometers. In yet further embodiments, the film thickness for each layer or sublayer can range from about 72 micrometers to about 320 micrometers.

(38) The lower barrier film **244** may have a greater thickness than the upper barrier film **242**, or vice versa. When thicker than the upper barrier film **242**, the lower barrier film **244** may be configured to provide a portion of the ground-contacting surface of the article of footwear **10**. Alternatively, the lower barrier film **244** and upper barrier film **242** may have equal thicknesses.

(39) One or both of barrier films **242** and **244** may independently be transparent, translucent, and/or opaque. For example, the upper barrier film **242** may be transparent, while the lower barrier film **244** is opaque. It is contemplated that upper barrier film **242** may be transparent or translucent, while lower barrier film **244** is opaque, or upper barrier film **242** may be opaque, while lower barrier film **244** is transparent or translucent, etc. As used herein, the term “transparent” for a barrier film and/or a fluid-filled chamber means that light passes through the barrier film in substantially straight lines and a viewer can see through the barrier film. In comparison, for an opaque barrier film, light does not pass through the barrier film and one cannot see clearly through the barrier film at all. A translucent barrier film falls between a transparent barrier film and an opaque barrier film, in that light passes through a translucent film but some of the light is scattered so that a viewer cannot see clearly through the .film.

(40) The chamber **246** may be produced from barrier films **242** and **244** using any suitable technique, such as thermoforming (e.g. vacuum thermoforming), blow molding, extrusion, injection molding, vacuum molding, rotary molding, transfer molding, pressure forming, heat sealing, casting, low-pressure casting, spin casting, reaction injection molding, radio frequency (RF) welding, and the like. In an aspect, barrier films **242** and **244** may be produced by co-extrusion followed by vacuum thermoforming to produce an inflatable chamber **246**, which may optionally include one or more valves (e.g., one way valves) that allows the chamber **246** to be filled with a fluid (e.g., gas).

(41) The chamber **246** may be provided in a fluid-filled (e.g., as provided in footwear **10**) or in an unfilled state. The chamber **246** may be filled to include any suitable fluid, such as a gas or liquid. In an aspect, the gas can include air, nitrogen (N.sub.2), oxygen gases (O.sub.2), inert gases, or any other suitable gas. In other aspects, the chamber **246** may alternatively include other media, such as pellets, beads, ground recycled material, and the like (e.g., foamed beads and/or rubber beads). The chamber **246** may be a membrane having a relatively low rate of transmittance of a fluid. When used alone or in combination with other materials in an airbag or bladder, the barrier membrane resiliently retains the fluid. Depending upon the structure and use of the airbag or bladder, the barrier membrane may retain the fluid at a pressure which is above, at, or below atmospheric pressure. The fluid provided to the chamber **246** may result in the chamber **246** being pressurized. The chamber **246** may have a pressure ranging from about atmospheric pressure to about 40 PSI. The chamber **246** may have a pressure ranging from about 10 PSI to about 25 PSI. The chamber **246** may have a pressure ranging from about 12 PSI to about 25 PSI. In an exemplary embodiment, the chamber **246** may have a pressure of about 15 PSI. It is contemplated that the chamber **246** may be pressurized to a pressure for desired characteristics of cushioning. Alternatively, the fluid provided to the chamber **246** may be at atmospheric pressure such that the chamber **246** is not pressurized but, rather, simply contains a volume of fluid at atmospheric pressure.

(42) The chamber **246** desirably may have a low gas transmission rate to preserve its retained gas

pressure. In some embodiments, the chamber **246** may have a gas transmission rate for nitrogen gas that is at least about ten (10) times lower than a nitrogen gas transmission rate for a butyl rubber layer of substantially the same dimensions. In an aspect, the chamber **246** may have a nitrogen gas transmission rate of 15 cubic-centimeter/square-meter.Math.atm.Math.day (cm.sup.3/m.sup.2.Math.atm.Math.day) or less for an average film thickness of 500 micrometers (based on thicknesses of barrier films **242** and **244**). In further aspects, the transmission rate is 10 cm.sup.3/m.sup.2.Math.atm.Math.day or less, 5 cm.sup.3/m.sup.2.Math.atm.Math.day or less, or 1 cm.sup.3/m.sup.2.Math.atm.Math.day or less.

(43) In some implementations, the upper barrier film **242** and the lower barrier film **244** may cooperate to define a geometry (e.g., thicknesses, width, and lengths) of the chamber **246**. For example, the web area **248** and the peripheral seam **250** may cooperate to bound and extend around the chamber **246** to seal the fluid (e.g., air) within the chamber **246**. Thus, the chamber **246** is associated with an area of the cushioning element **204** where interior surfaces of the upper and lower barrier films **242** and **244** are not joined together and, thus, are separated from one another. As shown in FIG. **4C**, a space formed between opposing interior surfaces of the upper and lower barrier films **242** and **244** defines an interior void of the chamber **246**. Similarly, exterior surfaces of the upper and lower barrier films **242** and **244** define an exterior profile of the chamber **246**. Accordingly, the upper and lower barrier films **242** and **244** define respective upper and lower surfaces of the cushioning element **204**.

(44) Referring to FIGS. **4A** and **4B**, the cushioning element **204** may include a medial tubular body **256**, a lateral tubular body **258**, one or more central tubular bodies **260**, and a valve **262**.

(45) Medial tubular body **256** and lateral tubular body **258** may be disposed on the respective medial side **24** and lateral side **22**. Medial tubular body **256** may extend from a medial first end **257** to a medial second end **261**. Lateral tubular body **258** may extend from a lateral first end **259** to a lateral second end **263**. Medial tubular body **256** may have a “J”-shape. Lateral tubular body **258** also may have a “J”-shape. Medial tubular body **256** and lateral tubular body **258** may be substantial mirrors of one another. The medial tubular body **256** may taper inward when approaching medial second end **261**. The lateral tubular body **258** may taper inward when approaching lateral second end **263**. The web areas **248a**, **248b**, **248c**, **248d**, and **248e** may be disposed between the medial tubular body **256** and the lateral tubular body **258**. The taper between the medial second end **261** and lateral second end **263**, the web area **248e**, and the peripheral seam **250** may form a pinch **269** in the heel most region of the cushioning element **240**. Pinch **269** may be substantially X-shaped when viewed from an exterior and on the side of the cushioning element **240**.

(46) The peripheral seam **250** may be disposed on a bottom surface of cushioning element **204**. Specifically, peripheral seam **250** may be disposed on a bottom surface of medial tubular body **256** and lateral tubular body **258**. A portion of peripheral seam **250** forming the pinch **269** may rise from the bottom surface of the medial tubular body **256** toward the medial second end **261**. A second portion of peripheral seam **250** forming the pinch **269** may rise from the bottom surface of the lateral tubular body **258** toward the lateral second end **263**. Peripheral seam **250** may not be visible through windows **232** and **232a**.

(47) The one or more central tubular bodies **260** may extend along a length of the cushioning element **240**. The one or more central tubular bodies **260** may span a width of the cushioning element **240** extending between the medial tubular body **256** and the lateral tubular body **258**. The one or more central tubular bodies **260** may taper radially inward at a medial end **265** as the one or more central tubular bodies **260** approach the medial tubular body **256**. In other words, the medial end **265** may form a reduced diameter section. The reduced diameter section comprises a diameter of the medial end **265** being less than a diameter of a portion of the medial tubular body **256**. Additionally, the one or more central tubular bodies **260** may taper inward at a lateral end **267** as the one or more central tubular bodies **260** approach the lateral tubular body **258**. In other words,

the lateral end **267** may form a reduced diameter section. The reduced diameter section comprises a diameter of the lateral end **267** being less than a diameter of a portion of the medial tubular body **258**. The web areas **248b**, **248c**, and **248d**, may be disposed between the one or more central tubular bodies **260**. The one or more central tubular bodies **260** may be in fluid communication with the medial tubular body **256** and the lateral tubular body **258**. The web area **248** may have disposed therein one or more pairs of holes **264**. The one or more pairs of holes **264** may be disposed at respective medial and lateral ends of the web area **248** corresponding to the medial and lateral ends **265** and **267**, respectively.

(48) Valve **262** may be present at the forefoot most area of the cushioning element **240** or any other suitable location for providing fluid to the cushioning element **204**. Valve **262** may be in fluid communication with the one or more central tubes **260**. The valve **262** may be configured to provide a fluid passage between a mold cavity (not shown) and the interior of the chamber **246**.

(49) With reference to FIGS. **4A-4C**, the web area **248** may formed at a bonded region of the upper barrier film **242** and the lower barrier film **244**, and may extend between and connect each of the medial tubular body **256**, the lateral tubular body **258**, and the one or more central tubular bodies **260**.

(50) Outsole

(51) With reference to FIG. **6**, outsole **206** may extend from the anterior end **18** to the posterior end **20**. Outsole **206** may have disposed thereon a plurality of traction elements **602a**, **602b**, **602c**, and **602d**. Outsole **206** may comprise a plurality of traction zones **604**, **606**, **608**, and **610**. Outsole **206** may further include one or more separators **612**. Separators **216** may be grooves in the outsole **206**. Alternatively, it is contemplated that separators **216** may be protrusions or the like extending from outsole **206** as necessary for dividing the outsole into multiple areas of traction for the article of footwear **10**. In an exemplary embodiment, separators **612** may be subdivided into separators **612a**, **612b**, and **612c**. It is contemplated that there may be one, two, three, or more separators as necessary to provide separation for various traction zones.

(52) Traction zone **604** may be disposed at the anterior end **18**. Traction zone **606** may be disposed adjacent to traction zone **604** and traction zone **608**. Traction zone **606** may be separated from traction zone **604** by separator **612a**. Traction zone **606** may be separated from traction zone **608** by separator **612b**. In other words, traction zone **606** may be bound at its top surface by traction zone **604** as well as separator **612a**, and traction zone **606** may be bound at its bottom surface by traction zone **608** as well as separator **612b**. Traction zone **608** may be disposed adjacent to traction zone **606** and traction zone **610**. Traction zone **608** may be separated from traction zone **606** by separator **612b**. Traction zone **608** may be separated from traction zone **610** by separator **612c**. In other words, traction zone **608** may be bound at its top surface by traction zone **606** as well as separator **612b**, and traction zone **608** may be bound at its bottom surface by traction zone **610** as well as separator **612c**. Traction zone **610** may be disposed adjacent to traction zone **608**, and separated by separator **612c**. Traction zone **610** may be disposed at the posterior end **20**. Traction zone **604** may not be disposed adjacent to either of traction zones **608** and **610**. Traction zone **606** may not be disposed adjacent to traction zone **610**. Traction zone **608** may not be disposed adjacent to transaction zone **604**. Traction zone **610** may not be disposed adjacent to either of traction zones **604** and **606**.

(53) Each of the traction zones **604**, **606**, **608**, and **610** may include the corresponding plurality of traction elements **602a**, **602b**, **602c**, and **602d**. The plurality of traction elements **602** may have a shape that is substantially square or rectangular. It is contemplated that the plurality of traction elements **602** may have a shape that is triangular, diamond, ovular, or irregular to provide a desired form of traction.

(54) The outsole **206** may be formed of materials, e.g., resilient materials among others, configured to impart properties of abrasion resistance and traction to the sole structure **102**. One or more of the outsole **102** and traction elements **602** may be formed of a first material having a higher durometer

than the other.

(55) Alternatively, the outsole **102** and the traction elements **602** may include the same material. The outsole **206** and the traction elements **602** may form the ground-engaging surface of the article of footwear **10**.

(56) The material of the outsole **206** can be a resilient material as described below. In one aspect, the resilient material is an elastomeric material comprising or consisting essentially of one or more elastomers. Examples of elastomers include butyl rubber, isoprene rubber, nitrile rubber, styrenic block copolymer rubber such as styrene-butadiene rubber, polyolefin rubber such as ethylene propylene rubber, silicone rubber, and combinations thereof. In some aspects, the one or more elastomers comprise thermoset elastomers. In other aspects, the one or more elastomers comprise thermoplastic elastomers. Examples of thermoplastic elastomers include thermoplastic styrenic block copolymer rubber, thermoplastic vulcanizate rubber, thermoplastic polyolefin rubber, thermoplastic polyurethane rubber, and combinations thereof. In addition to the one or more elastomers, the elastomeric material optionally can comprise one or more non-polymeric compounds, including fillers, processing aids, colorants, and combinations thereof.

(57) Assembled Sole Structure

(58) With reference to FIGS. 5A and 5B, the cushioning element **240** may be configured to cooperate with the internal cavity **224** of the midsole **202**. The cushioning element **240** may be disposed between the upper inner foam **223** and lower inner foam **225**. The upper and lower inner foams **223** and **225** may be in communication with the web area **248** of the cushioning element **240** thereby allowing the upper and lower inner foams **223** and **225** to surround the cushioning element **240**. Specifically, chamber **246** of the cushioning element **240** may be surrounded by the upper and lower inner foams **223** and **225** such that a viewer of the sole structure **102** would be able to see the environment on the opposite side due to the transparency of windows **232** and **232a** as well as barrier films **242** and **244** of the cushioning element **204**.

(59) When viewing the exterior of the footwear **10** from the ground up, the sole structure **102** may include the outsole **206**, the midsole **202**, and the cushioning element **204**.

(60) Referring to FIG. 5B, when viewing the stacked sole structure from the ground up along the exterior path of line A, the stacked sole structure may include the outsole **206** and the midsole **202**. When viewed along the exterior path of line B, the stacked sole structure may include the outsole **206**, the window **232**, cushioning element **204**, the chamber **246**, the upper and lower inner foams **223** and **225**, and the midsole **202**. The cushioning element **204**, the chamber **246**, and the upper and lower inner foams **223** and **225** are only visible through the window **232**. When viewed along the exterior path of line C, the stacked sole structure may include the outsole **206**, the window **232**, cushioning element **204**, web area **248**, the upper and lower inner foams **223** and **225**, and the midsole **202**. The cushioning element **204**, web area **248**, and the upper and lower inner foams **223** and **225** are only visible through the window **232**.

(61) Manufacturing Process

(62) In some implementations, the upper and lower barrier films **242** and **244** may be formed by respective mold portions. The respective mold portions may each define various surfaces for forming depressions and pinched surfaces corresponding to locations where the web area **248** and/or the peripheral seam **250** are formed when the upper barrier film **242** and the lower barrier film **244** are joined and bonded together. In some implementations, adhesive bonding may join the upper barrier film **242** and the lower barrier film **244** to form the web area **248** and the peripheral seam **250**. In other implementations, the upper barrier film **242** and the lower barrier film **244** may be joined to form the web area **248** and the peripheral seam **250** by thermal bonding. In some examples, one or both of the barrier films **242** and **244** may be heated to a temperature that facilitates shaping and melding. In some examples, the barrier films **242** and **244** may be heated prior to being located between their respective molds. In other examples, the mold may be heated to raise the temperature of the barrier films **242** and **244**. In some implementations, a molding

process used to form the fluid-filled chamber 246 may incorporate vacuum ports within mold portions to remove air such that the upper and lower barrier films 242 and 244 are drawn into contact with respective mold portions. In other implementations, fluids such as air may be injected into areas between the upper and lower barrier films 242 and 244 such that pressure increases cause the barrier films 242 and 244 to engage with surfaces of their respective mold portions.

(63) The multi-layered films may be produced by various means such as co-extrusion, lamination, layer-by-layer deposition, and the like. When co-extruding one or more barrier layers alone or with one or more second layers, selecting materials (e.g., a first barrier material and a second barrier material, or a single barrier material and a second material) having similar processing characteristics such as melt temperature and melt flow index, may reduce interlayer shear during the extrusion process, and may allow the alternating barrier layers and second layers to be co-extruded while retaining their structural integrities and desired layer thicknesses. In one example, the one or more barrier materials and optionally the second material when used, may be extruded into separate individual films, which may then be laminated together to form the multi-layered films.

(64) The multi-layered films may be produced using a layer-by-layer deposition process. A substrate, which optionally may comprise a second material or a barrier material, may be built into a multi-layered film by depositing a plurality of layers onto the substrate. The layers may include one or more barrier layers (e.g., first barrier layers, second barrier layers, etc.). Optionally, the layers may include one or more second layers. The one or more barrier layers and/or second layers may be deposited by any means known in the art such as, for example, dipping, spraying, coating, or another method. The one or more barrier layers may be applied using charged solutions or suspensions, e.g., cationic solutions or suspensions or anionic solutions or suspensions, including a charged polymer solution or suspension. The one or more barrier layers may be applied using a series of two or more solutions having opposite charges, e.g., by applying a cationic solution, followed by an anionic solution, followed by a cationic solution, followed by an anionic solution, etc.

(65) The barrier membranes, including the multi-layered films, have an overall thickness of from about 40 micrometers to about 500 micrometers, or about 50 micrometers to about 400 micrometers, or about 60 micrometers to about 350 micrometers. In one aspect, each individual layer of the plurality of layers of the multi-layered film has a thickness of from about 0.001 micrometers to about 10 micrometers. For example, the thickness of an individual barrier layer can range from about 0.001 micrometers to about 3 micrometers thick, or from about 0.5 micrometers to about 2 micrometers thick, or from about 0.5 micrometers to about 1 micrometer thick. The thickness of an individual second layer may range from about 2 micrometers to about 8 micrometers thick, or from about 2 micrometers to about 4 micrometers thick.

(66) In a further aspect, thickness of the films and/or their individual layers may be measured by any method known in the art such as, for example, ASTM E252, ASTM D6988, ASTM D8136, or using light microscopy or electron microscopy.

(67) In some aspects, the barrier membranes, including the multi-layered films, may have a Shore hardness of from about 35A to about 95A, optionally from about 55A to about 90A. In these aspects, hardness may be measured using ASTM D2240 using the Shore A scale.

(68) In one aspect, when a co-extrusion process is used to form the barrier membrane from a plurality of alternating barrier layers and second layers, the barrier material may have a melt flow index of from about 5 to about 7 grams per 10 minutes at 190 degrees Celsius when using a weight of 2.16 kilograms, while the second material may have a melt flow index of from about 20 to about 30 grams per 10 minutes at 190 degrees Celsius when using a weight of 2.16 kilograms. In a further aspect, the melt flow index of the barrier material may be from about 80 percent to about 120 percent of the melt flow index of the barrier material per 10 minutes when measured at 190 degrees Celsius when using a weight of 2.16 kilograms. In these aspects, melt flow index may be

measured using ASTM D1238. Alternatively or additionally, the barrier material or the second material or both may have a melting temperature of from about 165 degrees Celsius to about 183 degrees Celsius, or from about 155 degrees Celsius to about 165 degrees Celsius. In one such example, the barrier material may have a melting temperature of from about 165 degrees Celsius to about 183 degrees Celsius, while the second material has a melting temperature of from about 155 degrees Celsius to about 165 degrees Celsius. Further in these aspects, melting temperature can be measured using ASTM D3418.

(69) Materials

(70) Barrier films **242** and **244** can each be produced from an elastomeric material that includes one or more thermoplastic polymers and/or one or more cross-linkable polymers. In an aspect, the elastomeric material can include one or more thermoplastic elastomeric materials, such as one or more thermoplastic polyurethane (TPU) copolymers, one or more ethylene-vinyl alcohol (EVOH) copolymers, and the like.

(71) As used herein, “polyurethane” refers to a copolymer (including oligomers) that contains a urethane group (—N(C=O)O—). These polyurethanes can contain additional groups such as ester, ether, urea, allophanate, biuret, carbodiimide, oxazolidinyl, isocynaurate, uretdione, carbonate, and the like, in addition to urethane groups. In an aspect, one or more of the polyurethanes can be produced by polymerizing one or more isocyanates with one or more polyols to produce copolymer chains having (—N(C=O)O—) linkages.

(72) Examples of suitable isocyanates for producing the polyurethane copolymer chains include diisocyanates, such as aromatic diisocyanates, aliphatic diisocyanates, and combinations thereof. Examples of suitable aromatic diisocyanates include toluene diisocyanate (TDI), TDI adducts with trimethyloxypropane (TMP), methylene diphenyl diisocyanate (MDI), xylene diisocyanate (XDI), tetramethylxylene diisocyanate (TMXDI), hydrogenated xylene diisocyanate (HXDI), naphthalene 1,5-diisocyanate (NDI), 1,5-tetrahydronaphthalene diisocyanate, para-phenylene diisocyanate (PPDI), 3,3'-dimethyldiphenyl-4,4'-diisocyanate (DDDI), 4,4'-dibenzyl diisocyanate (DBDI), 4-chloro-1,3-phenylene diisocyanate, and combinations thereof. In some embodiments, the copolymer chains are substantially free of aromatic groups.

(73) In particular aspects, the polyurethane polymer chains are produced from diisocyanates including HMDI, TDI, MDI, H12 aliphatics, and combinations thereof. In an aspect, the thermoplastic TPU can include polyester-based TPU, polyether-based TPU, polycaprolactone-based TPU, polycarbonate-based TPU, polysiloxane-based TPU, or combinations thereof.

(74) In another aspect, the polymeric layer can be formed of one or more of the following: EVOH copolymers, poly(vinyl chloride), polyvinylidene polymers and copolymers (e.g., polyvinylidene chloride), polyamides (e.g., amorphous polyamides), amide-based copolymers, acrylonitrile polymers (e.g., acrylonitrile-methyl acrylate copolymers), polyethylene terephthalate, polyether imides, polyacrylic imides, and other polymeric materials known to have relatively low gas transmission rates. Blends of these materials as well as with the TPU copolymers described herein and optionally including combinations of polyimides and crystalline polymers, are also suitable.

(75) The barrier films **242** and **244** may include two or more sublayers (multilayer film) such as shown in Mitchell et al., U.S. Pat. No. 5,713,141 and Mitchell et al., U.S. Pat. No. 5,952,065, the disclosures of which are incorporated by reference in their entirety. In embodiments where the barrier films **242** and **244** include two or more sublayers, examples of suitable multilayer films include microlayer films, such as those disclosed in Bonk et al., U.S. Pat. No. 6,582,786, which is incorporated by reference in its entirety. In further embodiments, barrier films **242** and **244** may each independently include alternating sublayers of one or more TPU copolymer materials and one or more EVOH copolymer materials, where the total number of sublayers in each of barrier films **242** and **244** includes at least four (4) sublayers, at least ten (10) sublayers, at least twenty (20) sublayers, at least forty (40) sublayers, and/or at least sixty (60) sublayers.

(76) In one aspect, the barrier material may comprise or consist essentially of one or more

inorganic gas barrier compounds. The one or more inorganic gas barrier compounds can take the form of fibers, particulates, platelets, or combinations thereof. The fibers, particulates, platelets may comprise or consist essentially of nanoscale fibers, particulates, platelets, or combinations thereof. Examples of inorganic barrier compounds includes, for example, carbon fibers, glass fibers, glass flakes, silicas, silicates, calcium carbonate, clay, mica, talc, carbon black, particulate graphite, metallic flakes, and combinations thereof. The inorganic gas barrier component may comprise or consist essentially of one or more clays. Examples of suitable clays include bentonite, montmorillonite, kaolinite, and mixtures thereof. In one example, the inorganic gas barrier component consists of clay. Optionally, the barrier material may further comprise one or more additional ingredients, such as a polymer, processing aid, colorant, or any combination thereof. In aspects where the barrier material comprises or consists essentially of one or more inorganic barrier compounds, the barrier material may be described as comprising an inorganic gas barrier component consisting of all inorganic barrier compounds present in the barrier material. When one or more inorganic gas barrier compounds are included in the barrier material, the total concentration of the inorganic gas barrier component present in the barrier material may be less than 60 weight percent, or less than 40 weight percent, or less than 20 weight percent of the total composition. Alternatively, in other examples, the barrier material may consist essentially of the one or more inorganic gas barrier materials.

(77) In one aspect, the gas barrier compound comprises or consists essentially of one or more gas barrier polymers. The one or more gas barrier polymers can include thermoplastic polymers. In one example, the barrier material can comprise or consist essentially of one or more thermoplastic polymers, meaning that the barrier material comprises or consists essentially of a plurality of thermoplastic polymers, including thermoplastic polymers which are not gas barrier polymers. In another example, the barrier material comprises or consists essentially of one or more thermoplastic gas barrier polymers, meaning that all the polymers present in the barrier material are thermoplastic gas barrier polymers. The barrier material may be described as comprising a polymeric component consisting of all polymers present in the barrier material. For example, the polymeric component of the barrier material may consist of a single class of gas barrier polymer, such as, for example, one or more polyolefin, or can consist of a single type of gas barrier polymer, such as one or more ethylene-vinyl alcohol copolymers. Optionally, the barrier material may further comprise one or more non-polymeric additives, such as one or more filler, processing aid, colorant, or combination thereof.

(78) Many gas barrier polymers are known in the art. Examples of gas barrier polymers include vinyl polymers such as vinylidene chloride polymers, acrylic polymers such as acrylonitrile polymers, polyamides, epoxy polymers, amine polymers, polyolefins such as polyethylenes and polypropylenes, copolymers thereof, such as ethylene-vinyl alcohol copolymers, and mixtures thereof. Examples of thermoplastic gas barrier polymers include thermoplastic vinyl homopolymers and copolymers, thermoplastic acrylic homopolymers and copolymers, thermoplastic amine homopolymers and copolymers, thermoplastic polyolefin homopolymers and copolymers, and mixtures thereof. In one example, the one or more gas barrier polymers comprise or consist essentially of one or more thermoplastic polyethylene copolymers, such as, for example, one or more thermoplastic ethylene-vinyl alcohol copolymers. The one or more ethylene-vinyl alcohol copolymers can include from about 28 mole percent to about 44 mole percent ethylene content, or from about 32 mole percent to about 44 mole percent ethylene content. In yet another example, the one or more gas barrier polymers can comprise or consist essentially of one or more one or more polyethyleneimine, polyacrylic acid, polyethyleneoxide, polyacrylamide, polyamidoamine, or any combination thereof.

(79) The second material of the one or more second layers can comprise one or more polymers. Depending upon the class of gas barrier compounds used and the intended use of the multi-layered film, the second material may have a higher gas transmittance rate than the barrier material,

meaning that the second material is a poorer gas barrier than the barrier material. In some aspects, the one or more second layers may act as substrates for the one or more barrier layers, and may serve to increase the strength, elasticity, and/or durability of the multi-layered film. Alternatively or additionally, the one or more second layers may serve to decrease the amount of gas barrier material(s) needed, thereby reducing the overall material cost. Even when the second material has a relatively high gas transmittance rate, the presence of the one or more second layers, particularly when the one or more second layers are positioned between one or more barrier layers, may help maintain the overall barrier properties of the film by increasing the distance between cracks in the barrier layers, thereby increasing the distance gas molecules must travel between cracks in the barrier layers in order to pass through the multi-layered film. While small fractures or cracks in the barrier layers of a multi-layered film may not significantly impact the overall barrier properties of the film, using a larger number of thinner barrier layers can avoid or reduce visible cracking, crazing or hazing of the multi-layered film. The one or more second layers can include, but are not limited to, tie layers adhering two or more layers together, structural layers providing mechanical support to the multi-layered films, bonding layers providing a bonding material such as a hot melt adhesive material to the multi-layered film, and/or cap layers providing protection to an exterior surface of the multi-layered film.

(80) In some aspects, the second material is an elastomeric material comprising or consisting essentially of at least one elastomer. Many gas barrier compounds are brittle and/or relatively inflexible, and so the one or more barrier layers may be susceptible to cracking when subjected to repeated, excessive stress loads, such as those potentially generated during flexing and release of a multi-layered film. A multi-layered film which includes one or more barrier layers alternating with second layers of an elastomeric material results in a multi-layered film that is better able to withstand repeated flexing and release while maintaining its gas barrier properties, as compared to a film without the elastomeric second layers present.

(81) The second material comprises or consists essentially of one or more polymers. As used herein, the one or more polymers present in the second material are referred to herein as one or more “second polymers” or a “second polymer”, as these polymers are present in the second material. References to “second polymer(s)” are not intended to indicate that a “first polymer” is present, either in the second material, or in the multi-layered film as a whole, although, in many aspects, multiple classes or types of polymers are present. In one aspect, the second material comprises or consists essentially of one or more thermoplastic polymers. In another aspect, the second material comprises or consists essentially of one or more elastomeric polymers. In yet another aspect, the second material comprises or consists essentially of one or more thermoplastic elastomers. The second material can be described as comprising a polymeric component consisting of all polymers present in the second material. In one example, the polymeric component of the second material consists of one or more elastomers. Optionally, the second material can further comprise one or more non-polymeric additives, such as fillers, processing aids, and/or colorants.

(82) Many polymers which are suitable for use in the second material are known in the art. Exemplary polymers which can be included in the second material (e.g., second polymers) include polyolefins, polyamides, polycarbonates, polyimines, polyesters, polyacrylates, polyesters, polyethers, polystyrenes, polyureas, and polyurethanes, including homopolymers and copolymers thereof (e.g., polyolefin homopolymers, polyolefin copolymers, etc.), and combinations thereof. In one example, the second material comprises or consists essentially of one or more polymers chosen from polyolefins, polyamides, polyesters, polystyrenes, and polyurethanes, including homopolymers and copolymers thereof, and combinations thereof. In another example, the polymeric component of the second material consists of one or more thermoplastic polymers, or one or more elastomers or one or more thermoplastic elastomers, including thermoplastic vulcanizates. Alternatively, the one or more second polymers can include one or more thermoset or thermosettable elastomers, such as, for example, natural rubbers and synthetic rubbers, including

butadiene rubber, isoprene rubber, silicone rubber, and the like.

(83) Polyolefins are a class of polymers which include monomeric units derived from simple alkenes, such as ethylene, propylene and butene. Examples of thermoplastic polyolefins include polyethylene homopolymers, polypropylene homopolymers polypropylene copolymers (including polyethylene-polypropylene copolymers), polybutene, ethylene-octene copolymers, olefin block copolymers; propylene-butane copolymers, and combinations thereof, including blends of polyethylene homopolymers and polypropylene homopolymers. Examples of polyolefin elastomers include polyisobutylene elastomers, poly(alpha-olefin) elastomers, ethylene propylene elastomers, ethylene propylene diene monomer elastomers, and combinations thereof.

(84) Polyamides are a class of polymers which include monomeric units linked by amide bonds. Naturally-occurring polyamides include proteins such as wool and silk, and synthetic amides such as nylons and aramids. The one or more second polymers can include thermoplastic polyamides such as nylon 6, nylon 6-6, nylon-11, as well as thermoplastic polyamide copolymers.

(85) Polyesters are a class of polymers which include monomeric units derived from an ester functional group, and are commonly made by condensing dibasic acids such as, for example, terephthalic acid, with one or more polyols. In one example, the second material can comprise or consist essentially of one or more thermoplastic polyester elastomers. Examples of polyester polymers include homopolymers such as polyethylene terephthalate, polybutylene terephthalate, poly-1,4-cyclohexylene-dimethylene terephthalate, as well as copolymers such as polyester polyurethanes.

(86) Styrenic polymers are a class of polymers which include monomeric units derived from styrene. The one or more second polymers can comprise or consist essentially of styrenic homopolymers, styrenic random copolymers, styrenic block copolymers, or combinations thereof. Examples of styrenic polymers include styrenic block copolymers, such as acrylonitrile butadiene styrene block copolymers, styrene acrylonitrile block copolymers, styrene ethylene butylene styrene block copolymers, styrene ethylene butadiene styrene block copolymers, styrene ethylene propylene styrene block copolymers, styrene butadiene styrene block copolymers, and combinations thereof.

(87) Polyurethanes are a class of polymers which include monomeric units joined by carbamate linkages. Polyurethanes are most commonly formed by reacting a polyisocyanate (e.g., a diisocyanate or a triisocyanate) with a polyol (e.g., a diol or triol), optionally in the presence of a chain extender. The monomeric units derived from the polyisocyanate are often referred to as the hard segments of the polyurethane, while the monomeric units derived from the polyols are often referred to as the soft segments of the polyurethane. The hard segments can be derived from aliphatic polyisocyanates, or from organic isocyanates, or from a mixture of both. The soft segments can be derived from saturated polyols, or from unsaturated polyols such as polydiene polyols, or from a mixture of both. When the multi-layered film is to be bonded to natural or synthetic rubber, including soft segments derived from one or more polydiene polyols can facilitate bonding between the rubber and the film when the rubber and the film are crosslinked in contact with each other, such as in a vulcanization process.

(88) Examples of suitable polyisocyanates from which the hard segments of the polyurethane can be derived include hexamethylene diisocyanate (HDI), isophorone diisocyanate (IPDI), butylenediisocyanate (BDI), bisisocyanatocyclohexylmethane (HMDI), 2,2,4-trimethylhexamethylene diisocyanate (TMDI), bisisocyanatomethylcyclohexane, bisisochanatomethyltricyclodecane, norbornane diisocyanate (NDI), cyclohexane diisocyanate (CHDI), 4,4'-dicyclohexylmethane diisocyanate (H12MDI), diisocyanatododecane, lysine diisocyanate, toluene diisocyanate (TDI), TDI adducts with trimethylolpropane (TMP), methylene diphenyl diisocyanate (MDI), xylene diisocyanate (XDI), tetramethylxylene diisocyanate (TMXDI), hydrogenated xylene diisocyanate (HXDI), naphthalene 1,5-diisocyanate (NDI), 1,5-tetrahydronaphthalene diisocyanate, para-phenylene diisocyanate (PPDI), 3,3'-dimethyldiphenyl-

4,4'-diisocyanate (DDDI), 4,4'-dibenzyl diisocyanate (DBDI), 4-chloro-1,3-phenylene diisocyanate, and any combination thereof. In one aspect, the polyurethane comprises or consists essentially of hard segments derived from toluene diisocyanate (TDI), or from methylene diphenyl diisocyanate (MDI), or from both.

(89) The soft segments of the polyurethane can be derived from a wide variety of polyols, including polyester polyols, polyether polyols, polyester-ether polyols, polycarbonate polyols, polycaprolactone polyethers, and combinations thereof. In one aspect, the polyurethane comprises or consist essentially of monmeric units derived from C4-C12 polyols, or C6-C10 polyols, or C8 or lower polyols, meaning polyols with 4 to 12 carbon molecules, or with 6 to 10 carbon molecules, or with 8 or fewer carbon molecules in their chemical structures. In another aspect, the polyurethane comprises or consists essentially of monomeric units derived from polyester polyols, polyester-ether polyols, polyether polyols, and any combination thereof. In yet another aspect, the polyurethane comprises or consists essentially of soft segments derived from polyols or diols having polyester functional units. The soft segments derived from polyols or diols having polyester functional units can comprise about 10 to about 50, or about 20 to about 40, or about 30 weight percent of the soft segments present in the polyurethane.

(90) In an alternative embodiment, instead of being a fluid-filled bladder, the cushioning element **204** comprises a material, such as a foam or an unfoamed solid, to impart properties of cushioning, responsiveness, and energy distribution to the foot of the wearer. For example, the sole structure **102** may include the cushioning element **204**. The cushioning element **204** may comprise a foam. The foam may comprise a material. Example materials for the alternate cushioning element **204** may include those based on foaming or molding material, e.g. a resilient material, comprising one or more polymers, such as one or more elastomers (e.g., thermoplastic elastomers (TPE)). The one or more polymers may include aliphatic polymers, aromatic polymers, or mixtures of both; and may include homopolymers, copolymers (including terpolymers), or mixtures of both.

(91) In some aspects, the material is a thermoplastic material when it is combined with other elements to make the midsole, sole structure, or article of footwear, and remains thermoplastic, and is a thermoplastic material in the final product (e.g., in the finished midsole, finished sole structure or finished article of footwear). In other aspects, the material is a thermoplastic material when it is combined with other elements to make the midsole, sole structure, or article of footwear, and is subsequently cured into a crosslinked material in the manufacturing process, and thus is a crosslinked material in the final product. In yet other aspects, the material is crosslinked before being combined with other elements of the midsole, sole structure or article of footwear during the manufacturing process, and thus is a crosslinked material in the final product.

(92) In some aspects, the one or more polymers may include olefinic homopolymers, olefinic copolymers, or blends thereof. Examples of olefinic polymers include polyethylene, polypropylene, and combinations thereof. In other aspects, the one or more polymers may include one or more ethylene copolymers, such as, ethylene-vinyl acetate (EVA) copolymers, EVOH copolymers, ethylene-ethyl acrylate copolymers, ethylene-unsaturated mono-fatty acid copolymers, and combinations thereof.

(93) In further aspects, the one or more polymers may include one or more polyacrylates, such as polyacrylic acid, esters of polyacrylic acid, polyacrylonitrile, polyacrylic acetate, polymethyl acrylate, polyethyl acrylate, polybutyl acrylate, polymethyl methacrylate, and polyvinyl acetate; including derivatives thereof, copolymers thereof, and any combinations thereof.

(94) In yet further aspects, the one or more polymers may include one or more ionomeric polymers. In these aspects, the ionomeric polymers may include polymers with carboxylic acid functional groups, sulfonic acid functional groups, salts thereof (e.g., sodium, magnesium, potassium, etc.), and/or anhydrides thereof. For instance, the ionomeric polymer(s) may include one or more fatty acid-modified ionomeric polymers, polystyrene sulfonate, ethylene-methacrylic acid copolymers, and combinations thereof.

- (95) In further aspects, the one or more polymers may include one or more styrenic block copolymers, such as acrylonitrile butadiene styrene block copolymers, styrene acrylonitrile block copolymers, styrene ethylene butylene styrene block copolymers, styrene ethylene butadiene styrene block copolymers, styrene ethylene propylene styrene block copolymers, styrene butadiene styrene block copolymers, and combinations thereof.
- (96) In further aspects, the one or more polymers may include one or more polyamide copolymers (e.g., polyamide-polyether copolymers) and/or one or more polyurethanes (e.g., crosslinked polyurethanes and/or thermoplastic polyurethanes). Examples of suitable polyurethanes include those discussed above for barrier films **242** and **244**. Alternatively, the one or more polymers may include one or more natural and/or synthetic rubbers, such as polybutadiene and polyisoprene.
- (97) In one aspect, the resilient material is a foam. When the material is foamed, the resilient material may be foamed using a physical blowing agent which phase transitions to a gas based on a change in temperature and/or pressure, or a chemical blowing agent which forms a gas when heated above its activation temperature. For example, the chemical blowing agent may be an azo compound such as azodicarbonamide, sodium bicarbonate, and/or an isocyanate.
- (98) In some embodiments, the foam may be a crosslinked foam. In these embodiments, a peroxide-based crosslinking agent such as dicumyl peroxide may be used. Furthermore, the foam may include one or more fillers such as pigments, modified or natural clays, modified or unmodified synthetic clays, talc glass fiber, powdered glass, modified or natural silica, calcium carbonate, mica, paper, wood chips, and the like.
- (99) In another example, when the resilient material is a foam, the material may be foamed during a molding process, such as an injection molding process. A thermoplastic material may be melted in the barrel of an injection molding system and combined with a physical or chemical blowing agent and optionally a crosslinking agent (in order to make a crosslinked foam), and then injected into a mold under conditions which activate the blowing agent, forming a molded foam.
- (100) Optionally, when the resilient material is a foam, the foam may be a compression molded foam. Compression molding may be used to alter the physical properties (e.g., density, stiffness and/or durometer) of a foam, or to alter the physical appearance of the foam (e.g., to fuse two or more pieces of foam, to shape the foam, etc.), or both.
- (101) The compression molding process desirably starts by forming one or more foam preforms, such as by injection molding and foaming a material, e.g. a resilient material, by forming foamed particles or beads, by cutting foamed sheet stock, and the like. The compression molded foam may then be made by placing the one or more foam preforms in a compression mold, and applying sufficient pressure to the one or more preforms to compress the one or more foam preforms in a closed mold. Once the mold is closed, sufficient heat and/or pressure is applied to the one or more foam preforms in the closed mold for a sufficient duration of time to alter the foam preform(s), to form a skin on the outer surface of the compression molded foam, or to fuse individual foam particles to each other, to permanently or semi-permanently increase the density of the foam(s), or any combination thereof. Following the heating and/or application of pressure, the mold is opened and the molded foam article is removed from the mold.
- (102) In another aspect, the resilient material is an unfoamed solid. The material may be shaped using a molding process, including an injection molding process. In one example, when the material is an elastomeric material, the elastomeric material (e.g., uncured rubber) may be mixed in a Banbury mixer with an optional filler and a curing package such as, for example, a UV curing package or a thermal curing package including a sulfur-based or peroxide-based curing package, calendared, formed into shape, placed in a mold, and cured (e.g., using a UV curing process or a thermal curing process such as a vulcanization process).
- (103) The following clauses provide an exemplary configuration for an article of footwear and sole structure described above.
- (104) Clause 1. An article of footwear comprising: an upper; a sole structure, wherein the sole

structure includes: a midsole, wherein the midsole includes an enclosed cavity; a thermoformed cushioning element disposed within the enclosed cavity, wherein the cushioning element includes a first barrier film and a second barrier film enclosing an internal volume; an outsole layer, wherein the outsole layer forms a ground-engaging surface of the article of footwear.

(105) Clause 2. The article of footwear of clause 1, wherein the midsole includes a first window extending along a medial side of an exterior of the sole structure, and a second window extending along a lateral side of an exterior of the sole structure.

(106) Clause 3. The article of footwear of clause 2, wherein the midsole further includes a first foam and a second foam, wherein the cavity is defined by the first foam, the second foam, the first window, and the second window.

(107) Clause 4. The article of footwear of clause 3, wherein the first foam contacts a top surface of the cushioning element, and the second foam contacts a bottom surface of the cushioning element.

(108) Clause 5. The article of footwear of clause 2, wherein the cushioning element includes: a medial body extending from a first anterior medial end to a second posterior medial end, and disposed adjacent to the first window; a lateral body extending from a first anterior lateral end to a second posterior lateral end, and disposed adjacent to the second window; one or more central bodies disposed between the medial body and the lateral body extending in a medial direction and a lateral direction; and a web area interconnecting disposed between the medial body, the lateral body, and the one or more central bodies.

(109) Clause 6. The article of footwear of clause 5, wherein the medial body, the lateral body, and the one or more central bodies, are fluid-filled.

(110) Clause 7. The article of footwear of clause 6, wherein the medial body, the lateral body, and the one or more central bodies, are all in fluid communication with one another.

(111) Clause 8. The article of footwear of clause 5, wherein the first barrier film and the second barrier film are coupled to one another at a peripheral seam disposed around an exterior of the medial body and the lateral body of the cushioning element.

(112) Clause 9. The article of footwear of clause 8, wherein no portion of the peripheral seam is visible from the exterior of the article of footwear, through either the lateral window or the medial window.

(113) Clause 10. The article of footwear of clause 5, wherein the first barrier film and the second barrier film are also coupled to one another at the web area disposed between the medial body, the lateral body, and the one or more central bodies, wherein the web area is positioned in a plane offset from a plane containing substantially all of the peripheral seam.

(114) Clause 11. The article of footwear of clause 5, wherein the only portions of the web area visible from exterior of the article of footwear and through either 1) the first window or the second window, are co-planar to one another.

(115) Clause 12. The article of footwear of clause 2, wherein the first window and the second window each include a transparent material that is distinct from the cushioning element.

(116) Clause 13. The article of footwear of clause 5, wherein each of the one or more central bodies is coupled to the medial body at a reduced diameter section.

(117) Clause 14. The article of footwear of clause 13, wherein each of the one or more central bodies is coupled to the lateral body at a reduced diameter section.

(118) Clause 15. The article of footwear of clause 5, wherein the medial body and the lateral body are substantially J-shaped.

(119) Clause 16. The article of footwear of clause 5, wherein the medial body and the lateral body are mirror images of one another.

(120) Clause 17. The article of footwear of clause 5, wherein the one or more central bodies includes exactly four central bodies.

(121) Clause 18. The article of footwear of clause 5, wherein a hollow interior of each of the one or more central bodies is visible from exterior of the article of footwear, through each of the first

window and the second window.

(122) Clause 19. An article of footwear comprising: an upper; a sole structure, wherein the sole structure includes: a midsole, wherein the midsole includes an enclosed cavity; a cushioning element disposed within the enclosed cavity, wherein the cushioning element includes a first barrier film and a second barrier film enclosing an internal volume, wherein the first barrier film and the second barrier film are coupled to one another at a peripheral seam that extends along an outer surface of the cushioning element; a first window extending along a medial side of the sole structure; a second window extending along a lateral side of the sole structure, wherein the peripheral seam is positioned below bottom surfaces of the first window and the second window; and an outsole layer, wherein the outsole layer forms a ground-engaging surface of the article of footwear.

(123) Clause 20. A cushioning element for an article of footwear, the cushioning element comprising: a first barrier film and a second barrier film enclosing an internal volume, a medial body extending from a first anterior medial end to a second posterior medial end; a lateral body extending from a first anterior lateral end to a second posterior lateral end; one or more central bodies disposed between the medial body and the lateral body extending in a medial direction and a lateral direction; and wherein the first barrier film and the second barrier film are coupled to one another at 1) a peripheral seam that extends along an outer surface of the cushioning element, the peripheral seam being positioned substantially in a first plane, and 2) at a web area disposed between the medial body, the lateral body, and the one or more central bodies, wherein the web area is disposed in a second plane that is offset from the first plane.

(124) Clause 21. The cushioning element of clause 20, wherein the first barrier film and the second barrier film each includes a multi-layer film.

(125) Clause 22. The cushioning element of clause 20, wherein the medial body, the lateral body, and the one or more central bodies, are fluid-filled.

(126) Clause 23. The cushioning element of clause 22, wherein the medial body, the lateral body, and the one or more central bodies, are all in fluid communication with one another.

(127) Clause 24. The cushioning element of clause 20, wherein the first barrier film and the second barrier film are coupled to one another at a peripheral seam disposed around an exterior of the medial body and the lateral body of the cushioning element.

(128) Clause 25. The cushioning element of clause 20, wherein the first barrier film and the second barrier film are also coupled to one another at the web area disposed between the medial body, the lateral body, and the one or more central bodies, wherein the web area is positioned in a plane offset from a plane containing substantially all of the peripheral seam.

(129) Clause 26. The cushioning element of clause 20, wherein each of the one or more central bodies is coupled to the medial body at a reduced diameter section, the reduced diameter section being an area with a diameter of a medial end of the one or more central bodies being less than a diameter of a portion of the medial body.

(130) Clause 27. The cushioning element of clause 26, wherein each of the one or more central bodies is coupled to the lateral body at a reduced diameter section, the reduced diameter section being an area with a diameter of a lateral end of the one or more central bodies being less than a diameter of a portion of the lateral body.

(131) Clause 28. The cushioning element of clause 20, wherein the medial body and the lateral body are substantially J-shaped.

(132) Clause 29. The cushioning element of clause 20, wherein the medial body and the lateral body are mirror images of one another.

(133) Clause 30. The cushioning element of clause 20, wherein the one or more central bodies includes exactly four central bodies.

(134) Clause 31. The cushioning element of clause 21, wherein the multi-layer film includes a plurality of layers.

- (135) Clause 32. The cushioning element of clause 31, wherein the plurality of layers ranges from 5 layers to 400 layers.
- (136) Clause 33. The cushioning element of clause 32, wherein each of the plurality of layers has a thickness between 40 micrometers to 500 micrometers.
- (137) Clause 34. A sole structure comprising the cushioning element according to any one of clauses 20-33.
- (138) Clause 35. An article of footwear comprising the sole structure of clause 34.
- (139) Clause 36. A method of manufacturing a cushioning element, the method comprising: molding a first barrier film and a second barrier film via one or more mold portions; bonding the first barrier film and the second barrier to form a web area and a peripheral seam, wherein the web area is positioned in a plane offset from a plane containing substantially all of the peripheral seam; heating the first barrier film and the second barrier film; and forcing the first barrier film and the second barrier film into contact with the one or more mold portions.
- (140) Clause 37. A cushioning element manufactured by the method of clause 36.
- (141) Clause 38. A sole structure manufactured by coupling the cushioning element of clause 37 with a midsole.
- (142) Clause 39. An article of footwear manufactured by coupling the sole structure of clause 38 with an upper.

Claims

1. An article of footwear comprising: an upper; a sole structure, wherein the sole structure includes: a midsole, wherein the midsole includes an enclosed cavity and at least one window extending between top and bottom planes that are offset from one another by a gap; a thermoformed cushioning element disposed within the enclosed cavity, wherein the cushioning element includes a first barrier film and a second barrier film coupled to one another at a peripheral seam enclosing an internal volume, the peripheral seam formed at a side of the thermoformed cushioning element which is proximate the at least one window, wherein the peripheral seam is disposed outside of the gap such that the peripheral seam is not visible through the at least one window; and an outsole layer, wherein the outsole layer forms a ground-engaging surface of the article of footwear.
2. The article of footwear of claim 1, wherein the midsole includes a first window extending along a medial side of an exterior of the sole structure, and a second window extending along a lateral side of an exterior of the sole structure.
3. The article of footwear of claim 2, wherein the midsole further includes an first foam and a second foam, wherein the cavity is defined by the first foam, the second foam, the first window, and the second window.
4. The article of footwear of claim 3, wherein the first foam contacts an outer surface of the cushioning element, and the second foam contacts a different part of the outer surface of the cushioning element than the first foam.
5. The article of footwear of claim 2, wherein the cushioning element includes: a medial body extending from a first anterior medial end to a second posterior medial end, and disposed adjacent to the first window; a lateral body extending from a first anterior lateral end to a second posterior lateral end, and disposed adjacent to the second window; one or more central bodies disposed between the medial body and the lateral body extending in a medial direction and a lateral direction; and a web area disposed between the medial body, the lateral body, and the one or more central bodies.
6. The article of footwear of claim 5, wherein the medial body, the lateral body, and the one or more central bodies, are fluid-filled.
7. The article of footwear of claim 6, wherein the medial body, the lateral body, and the one or more central bodies, are all in fluid communication with one another.

8. The article of footwear of claim 5, wherein the first barrier film and the second barrier film are coupled to one another at a peripheral seam disposed around an exterior of the medial body and the lateral body of the cushioning element.
9. The article of footwear of claim 8, wherein no portion of the peripheral seam is visible from the exterior of the article of footwear, through either the lateral window or the medial window.
10. The article of footwear of claim 5, wherein the first barrier film and the second barrier film are also coupled to one another at the web area disposed between the medial body, the lateral body, and the one or more central bodies, wherein the web area is positioned in a plane offset from a plane containing substantially all of the peripheral seam.
11. The article of footwear of claim 5, wherein the only portions of the web area visible from exterior of the article of footwear and through either the first window or the second window are coplanar to one another.
12. The article of footwear of claim 5, wherein each of the one or more central bodies is coupled to the medial body at a reduced diameter section.
13. The article of footwear of claim 12, wherein each of the one or more central bodies is coupled to the lateral body at a reduced diameter section.
14. The article of footwear of claim 5, wherein the medial body and the lateral body are substantially J-shaped.
15. The article of footwear of claim 5, wherein the medial body and the lateral body are mirror images of one another.
16. The article of footwear of claim 5, wherein the one or more central bodies includes exactly four central bodies.
17. The article of footwear of claim 5, wherein a hollow interior of each of the one or more central bodies is visible from exterior of the article of footwear, through each of the first window and the second window.
18. The article of footwear of claim 2, wherein the first window and the second window each include a transparent material that is distinct from the cushioning element.
19. An article of footwear comprising: an upper; a sole structure, wherein the sole structure includes: a midsole, wherein the midsole includes an enclosed cavity; a cushioning element disposed within the enclosed cavity, wherein the cushioning element includes a first barrier film and a second barrier film enclosing an internal volume, wherein the first barrier film and the second barrier film are coupled to one another at a peripheral seam that extends along a side surface of the cushioning element; a first window extending along a medial side of the sole structure; a second window extending along a lateral side of the sole structure, wherein the peripheral seam is positioned below bottom surfaces of the first window and the second window, wherein the side surface of the cushioning element is proximate the first window; and an outsole layer, wherein the outsole layer forms a ground-engaging surface of the article of footwear.
20. A cushioning element for an article of footwear, the cushioning element comprising: a first barrier film and a second barrier film enclosing an internal volume; a medial body extending from a first anterior medial end to a second posterior medial end; a lateral body extending from a first anterior lateral end to a second posterior lateral end; one or more central bodies disposed between the medial body and the lateral body extending in a medial direction and a lateral direction; and wherein the first barrier film and the second barrier film are coupled to one another at 1) a peripheral seam that extends along a side surface of the cushioning element, the peripheral seam including a first substantially flat section positioned substantially in a first plane, a second substantially flat section positioned in a second plane offset from the first plane, and an angled section positioned between the first substantially flat section and the second substantially flat section and 2) at a web area disposed between the medial body, the lateral body, and the one or more central bodies, wherein the web area is disposed in a second plane that is offset from the first plane.

